MIL-C-7907A(WP)
3 May 1965

Superseding
MIL-C-7907(Aer)
31 July 1952

MILITARY SPECIFICATION

CLEANING COMPOUND, DECONTAMINATING (FOR SOILED AND RADIOACTIVE, CONTAMINATED SURFACES)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification establishes the formulation and requirements for a decontaminating cleaning compound that is soluble in water, hard water and sea water (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-C-105	Calcium Chloride, Dihydrate; and Calcium Chloride, Anhydrous; Technical
O-S-635	Sodium Polyphosphate, Technical, Water Treatment

FSC 6850

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Federal (Cont'd)

O-S-642	Sodium Phosphate, Tribasic, Technical; Anhydrous, Dodecahydrate, and Monohydrate
O-S-377	Sodium Pyrophosphate, Anhydrous, Technical (Tetrasodiumpyrophosphate)
QQ-A-250/4	Aluminum Alloy, 2024, Plate and Sheet
QQ-A-250/5	Aluminum Alloy, Alclad 2024, Plate and Sheet
QQ-C-502	Copper Rods and Shapes, and Flat Products with Finished Edges (Flat Wire, Strips and Bars)
QQ-M-44	Magnesium Alloy, Plate and Sheet (AZ31)
QQ-P-416	Plating, Cadmium (Electrodeposited)
SS-S-535	Sodium Borate, Decahydrate, Technical (Borax)
SS-S-550	Sodium Chloride, Technical, for Water Softening Units
TT-N-95	Naphtha, Aliphatic
TT-P-343	Pigment, Carbon-Black, Dry
TT-T-291	Thinner; Paint, Volatile Mineral Spirits (Petroleum Spirits)
PPP-C-301	Chemicals, Dry and Paste, Packaging and Packing of

Military

MIL-M-3171	Magnesium Alloy, Process for Corrosion Protection of
MIL-C-5410	Cleaning Compound, Aluminum Surface, Non-Flame-Sustaining
MIL-C-5425	Plastic Sheet, Acrylic, Heat Resistant

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Military (Cont'd)

MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-L-6082	Lubricating Oil, Aircraft Reciprocating Engine (Piston)
MIL-S-6721	Steel, Corrosion and Heat Resistant (Chemically Stabilized) Plate, Sheet and Strip
MIL-S-7952	Steel, Sheet and Strip, Uncoated, Carbon (1020 and 1025) Aircraft Quality
MIL-R-7962	Primer Coating, Cellulose Nitrate Modified Alkyd Type, Corrosion-Inhibiting, Fast Drying (For Spray Application Over Pretreatment Coating)
MIL-C-8514	Coating Compound, Metal Pretreatment, Resin-Acid
MIL-C-18687	Cleaning Compound, Biodegradeable
MIL-L-19537	Lacquer, Acrylic-Nitrocellulose, Gloss (For Aircraft Use)
MIL-C-22750	Coating, Epoxy-Ployamide, Chemical and Solvent Resistant, for Weapons Systems
MIL-P-23377	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant, for Weapons Systems
MIL-S-50004	Sodium Sulfate, Anhydrous, Technical (For Ammunition Use)
MIL-S-51078	Sodium Hexametaphosphate, Technical

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Data - No data is required by this specification or by the applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2 Material - The manufacturer shall certify by affidavit that the cleaning compound conforms to the requirements and proportions specified in Table I (see 4.4.1).

TABLE I

Ingredient	Specification	Percent by weight
Sodium pyrophosphate (anhydrous)	O-S-377	10
Sodium tripolyphosphate (anhydrous)	O-S-635 (Type III)	30
Sodium phosphate, tribasic (anhydrous)	O-S-642	20
Non-ionic detergent	MIL-C-18687(Type II)	10
Methylcellulose 1/	33-56 percent active ingredient	5
Sodium xylenesulfonate 2/	--	5
Sodium Hexametaphosphate	MIL-S-51078	10
Borax	SS-S-535	10
		100

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1/ Equivalent to Methocel CAM, manufactured by Dow Chemical Company, Midland, Michigan.

2/ Equivalent to Naxonate G, manufactured by Wyandotte,Chemical Corporation, Wyandotte, Michigan.

3.2.1 Preparation of compound - The powdered ingredients shall be thoroughly blended together. The non-ionic detergent shall be added very slowly during the blending operation.

3.3 Solubility in distilled water - When tested as specified in 4.5.1, the cleaning compound shall dissolve readily in distilled water producing no more than a slight cloudiness. If the extent of the cloudiness is questionable; the volume of sediment, after centrifuging, shall be not more than 0.05 percent.

3.4 Stability in distilled water - After being subjected to temperatures of 4°C (39.2°F) and 98°C (208.4°F) for 10 minute periods as specified in 4.5.2, the cleaning compound-distilled water solution shall show no sign of separation, sedimentation, gelling or noticeable change in viscosity and only a slight cloudiness. If the extent of cloudiness is questionable; the volume of sediment, after centrifuging, shall be not more than 0.05 percent.

3.5 Solubility in synthetic hard water - When tested as specified in 4.5.3, the volume of sediment in the cleaning compound-synthetic hard water solution shall be not more than 0.8 percent.

3.6 Solubility in synthetic sea water - When tested as specified in 4.5.4, the volume of sediment in the cleaning compound-synthetic sea water solution shall be not more than 7.0 percent.

3.7 pH - When tested as specified in 4.5.5, the pH of the cleaning compound-distilled water solution shall be not less than 8.0 nor more than 10.0.

3.8 Total Alkalinity - When tested as specified in 4.5.6, the total alkalinity of the cleaning compound, calculated as Na_2O, shall be not more than 21 percent.

3.9 Surface tension - When tested as specified in 4.5.7, the surface tension of the cleaning compound test solutions shall conform to the requirements of Table II.

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TABLE II

Cleaning compound dissolved in	Surface tension, maximum (in dynes per centimeter)
distilled water	34
synthetic hard water	32
synthetic sea water	32

3.10 Interfacial tension - When tested as specified in 4.5.8, the interfacial tension between a layer of white paraffin oil and the specified cleaning compound test solution shall conform to the requirements of Table III.

TABLE III

Cleaning compound dissolved in	Interfacial tension, maximum, (in dynes per centimeter)
distilled water	4
synthetic hard water	4
synthetic sea water	3

3.11 Radioactive soil removal - When tested as specified in 4.5.9, the cleaning compound-distilled water solution shall remove not less than 20 percent of the radioactivity from the soiled retaining plugs.

3.12 Cleaning ability - When tested as specified in 4.5.10, the cleaning compound test solutions shall completely remove an oil-carbon mixture from the test panels.

3.13 Rinsing ability - When tested as specified in 4.5.11, the cleaning compound shall be completely removed from the test panels after rinsing with distilled water, synthetic hard water or synthetic sea water; except that a slight film on the panels after rinsing with synthetic hard water shall be acceptable.

3.14 Corrosiveness - When tested as specified in 4.5.12, the cleaning compound-distilled water solution shall cause no discoloration, corrosion, chemical attack or any other deterioration of the test panels; except that a slight streaking of the aluminum alloy panel and a slight dulling of the alclad aluminum alloy panel shall be acceptable.

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3.15 Effect on painted surfaces - When tested as specified in 4.5.13, the cleaning compound test solutions shall cause no softening, blistering, discoloration, cracking or any other deterioration of the epoxy coated or lacquered test panels.

3.16 Effect on acrylic base plastics - When tested as specified in 4.5.14, the cleaning compound test solutions shall cause no crazing, chemical attack, or any other deterioration of the stressed acrylic panels.

3.17 Workmanship - When examined visually, the cleaning compound shall be a homogeneous mixture that is free from lumps and foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility - Unless otherwise specified by the Government, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the examinations and tests set forth in the specification where such are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot formation - For purposes of sampling, a lot shall consist of a manufacturer's batch. If the material cannot be identified by batch, a lot shall consist of not more than 10,000 pounds of cleaning compound. A batch is defined as the end product of all raw materials mixed or blended in a single operation.

4.3 Lot acceptance samples -

* 4.3.1 Lot acceptance samples for tests - Two five pound samples of cleaning compound shall be selected at random from each lot, either during or after the filling operation.

* 4.3.2 Lot acceptance samples of preparation for delivery - A quantity of filled containers and shipping containers fully prepared for delivery shall be taken at random from each lot in accordance with Standard MIL-STD-105 at inspection level I.

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* 4.4 Lot acceptance inspections -

* 4.4.1 Lot acceptance inspection of samples for tests - Each sample selected in accordance with 4.3.1 shall be examined and tested to determine conformance to all the requirements of Section 3 of this specification except 3.2. The manufacturer shall certify by affidavit that the cleaning compound conforms to the materials requirements set forth in 3.2. Non-conformance of a sample to a requirement shall be cause for rejection of the lot represented by the sample.

* 4.4.2 Lot acceptance inspection of Preparation for Delivery - Each lot acceptance sample of a filled container or of a shipping container fully prepared for delivery (selected in accordance with 4.3.2) shall be considered a unit of product for purposes of this inspection. Each sample shall be visually examined for defects of the containers and closures for evidence of leakage, for unsatisfactory marking, for net contents and to determine compliance with all the other requirements of Section 5 of this specification. If the number of defective units of product in the lot sample exceeds the acceptable quality level (AQL) of 2.5 percent, the lot represented by the sample shall be rejected.

4.5 Test methods - Unless otherwise specified herein, all tests shall be performed at a laboratory temperature of 75 ± 2°F (23.9 ± 1.1°C) and a relative humidity of 50 ± 4 percent.

4.5.1 Solubility in distilled water - Add 43 grams of the cleaning compound to 6 liters of distilled water and mix thoroughly. After allowing to stand undisturbed for 5 minutes, observe the solution for cloudiness. If the extent of cloudiness is questionable, the volume of sediment shall be determined as specified in 4.5.1.1.

4.5.1.1 Sediment - One hundred ml of the cleaning compound test solution shall be added to each of the two 100 ml graduated centrifuge tubes, graduated for reading to 0.1 ml. Then the tubes shall be centrifuged at a speed of 1500-1800 rpm for 15 minute periods until the volume of sediment in each tube becomes constant. The sediment volumes shall be read to 0.1 ml and the average of the two results shall be reported. The diameter of swing (tip to tip of the whirling tubes) of the centrifuge shall be between 16 and 18 inches.

4.5.2 Stability in distilled water - Add 50 milliters of the cleaning compound-distilled water solution specified in 4.5.1 to each of two test tubes. Cool one tube to 4° ± 1°C (39 ± 2°F) and heat the other to 98 ± 1°C (208 ± 2°F). After 10 minutes at these temperatures,

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condition at room temperature for 18 hours and then observe each tube for any evidence of separation, sedimentation, gelling, noticeable change in viscosity or cloudiness. If the extent of cloudiness is questionable, the volume of sediment shall be determined as specified in 4.5.1.1.

4.5.3 Solubility in synthetic hard water-- Prepare the synthetic hard water by dissolving 3.0 grams of calcium acetate, $Ca(CH_3COO)_2 \cdot H_2O$ and 2.1 grams of magnesium sulfate, $MgSO_4 \cdot 7H_2O$ (Technical Grades) in 6 liters of distilled water. Add 43 grams of the cleaning compound to 6 liters of the synthetic hard water, and mix thoroughly. After allowing to remain undisturbed for 5 minutes, determine the volume of sediment as specified in 4.5.1.1.

4.5.4 Solubility in synthetic sea water - The synthetic sea water shall be prepared in accordance with Table IV. Add 43 grams of the cleaning compound to 6 liters of the synthetic sea water and mix thoroughly. Allow the solution to stand undisturbed for 5 minutes. The volume of sediment in the test solution shall then be determined as specified in 4.5.1.1.

TABLE IV

Ingredient	Specification	Quantity in grams
Sodium Chloride	SS-S-550	150.0
Magnesium Chloride ($MgCL_2 \cdot 6H_2O$)	Technical Grade	66.0
Calcium Chloride, dihydrate	O-C-105	9.6
Sodium Sulfate, Anhydrous	MIL-S-50004	24.0
Water	Add distilled water to make a total volume of 6 liters.	

4.5.5 pH - The cleaning compound-distilled water test solution specified in 4.5.1 shall be tested with a pH electrometer having a sensitivity and readability of 0.05. The meter shall have a sealed glass type electrode and a potassium chloride-calomel reference electrode. Calibrate the meter against a standard buffer solution prior to determining the pH of the test solution.

4.5.6 Total alkalinity -

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4.5.6.1 Indicator solution - Grind a 0.1 gram sample of bromophenol blue in a mortar with 1.5 milliliters (ml) of 0.1 normal NaOH. Transfer this mixture to a volumetric flask, dilute to 250 ml with distilled water; adjust to neutrality.

4.5.6.2 Procedure - Add 5 drops of the indicator solution prepared in 4.5.6.1 to a 100 ml aliquot of the cleaning compound-distilled water solution specified in 4.5.1. A blue coloration should develop. Titrate the solution with 0.5 normal hydrochloric or sulfuric acid to the first appearance of yellow. Calculate the total alkalinity as follows:

$$\text{Percent total Alkalinity (as } Na_2O) = \frac{3.1\ (A)\ (N)}{0.720}$$

where A = ml of acid
N = normality of acid

4.5.7 Surface tension -

4.5.7.1 Apparatus and method - An instrument equivalent to the Cenco-Du Nuoy tensiometer (manufactured by Central Scientific Company, 1702 Irving Park, Chicago, Illinois) shall be used. The instrument shall be equipped with a platinum-iridium ring having a circumference of 6 centimeters, and shall be capable of reading to 0.1 dyne per cm with a precision of ± 0.05 dyne per cm. Level the instrument and calibrate it against standard weights and pure chemicals. Thoroughly clean the ring and glassware and adjust the instrument so that the ring system is in its zero position when the ring is dry and the scale reading is zero. Follow the recommended directions supplied by the manufacturer.

4.5.7.2 Measurements - Surface tension measurements shall be made on a 50 ml portion of each of the cleaning compound test solutions specified in 4.5.1, 4.5.3 and 4.5.4.

4.5.8 Interfacial tension - Using the apparatus and method specified in 4.5.7.1, interfacial tension measurements shall be made after a portion of each of the cleaning compound test solutions specified in 4.5.1, 4.5.3 and 4.5.4 have been covered with an approximately equal volume of U.S.P. grade white paraffin oil.

4.5.9 Radioactive soil removal -

4.5.9.1 Radioactive soil - A partial formulation of the radioactive soil shall be made as specified in Table V. Heat the mixture in a porcelain crucible at a temperature of 1000 to 1200°C (1832 to 2192°F) for 3 hours to convert the ingredient metals to their oxides. Cool to

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room temperature and transfer to a stainless steel ball mill that is one half filled with stainless steel balls. Add 50 grams of uranium oxide and 70 milliliters (ml) of distilled water to the mill also. (Uranium oxide can be prepared by heating uranium acetate at 1000 to 1200°C (1832 to 2192°F) in an electric furnace.) After letting the mill run for 3 hours, transfer the mixture to a beaker. Rinse the mill and balls with distilled water, adding the washings to the beaker. Continue rinsing the mill until the washings, after evaporation, exhibit no appreciable radiation as measured by a Geiger-Müller counter. Reduce the volume of the soil slurry to 1 liter by evaporation. Add 50 grams of uranyl nitrate to the slurry and mix thoroughly.

TABLE V
Radioactive Soil (Partial formulation)

INGREDIENT	WEIGHT IN GRAMS
Lanthanum Nitrate	16.93
Molybdenum Oxide	16.40
Ferric Oxide	11.11
Nickel Chloride	0.08
Mercuric Oxide	0.03
Stannous Chloride	0.38
Tellurium Dioxide	7.41
Antimony Trioxide	36.48
Cerium Oxide	10.69
Arsenic Oxide	0.33
Cobalt Nitrate	0.08
Lead Oxide	0.08
Zinc Oxide	Trace
Selenious Acid	Trace

4.5.9.2 Soil retaining plugs - Soil retaining plugs shall be constructed from Alclad 2024 aluminum alloy conforming to QQ-A-250/5 and be equivalent to the E-7B one inch ring and disc assembly manufactured by Tracerlab Inc., 1601 Trapelo Rd; Waltham, Mass. The plugs shall be cleaned, chemically treated and coated with the lacquer system as specified in 4.5.13.1.

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4.5.9.3 Application of radioactive soil - Place the cylinder portion of the Tracerlab Inc. E-8B precipitation apparatus or a similar tapered stainless steel cylinder over each of five soil retaining plugs prepared as specified in 4.5.9.2. The radioactive soil slurry prepared in 4.5.9.1 shall be then shaken vigorously and a small portion of it transferred to a graduated cylinder fitted with a ground glass stopper. Dilute the slurry with 4 times its volume of distilled water. After shaking vigorously, transfer 0.5 ml of the diluted slurry to each plug by means of a pipet. Evaporate the water from each plug using an infra-red lamp under a hood. Remove the cylinder from the plugs after the major portion of the water has been evaporated, and continue heating until the plugs are dry.

4.5.9.4 Procedure - Place the five test plugs containing the dry radioactive soil (that were prepared in 4.5.9.3) in a liter beaker with 700 ml of distilled water. Position a glass-rod attached to an electric stirrer in the beaker so that its end is 1 inch above the plugs. Stir the contents of the beaker for 5 minutes with the speed of the stirrer adjusted so that the plugs do not rotate. Remove the plugs, wash off in a strong stream of distilled water from a wash bottle, and dry in a fume hood under an infra-red heat lamp. Radioactivity measurements (R_b) shall then be made on each plug using an end window tube Geiger-Müller Counter. After the measurements have been taken place the five plugs in a one liter beaker with 700 ml of the cleaning compound-distilled water solution specified in 4.5.1. Arrange the stirrer as described above and stir for 30 minutes. Remove the plugs and rinse in a strong stream of distilled water from a wash bottle. Dry in a hood under an infra-red lamp. Radioactivity measurements (R_a) shall then be taken on each plug. Calculations shall be made as follows:

$$\text{Percent Radioactivity Removal} = \frac{R_b - R_a}{R_b} (100)$$

where R_b = Radiation count, corrected for background, after cleaning with distilled water.

R_a = Radiation count, corrected for background, after cleaning with the cleaning compound-distilled water solution.

Note: The radiation count shall be recorded as counts per second.

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4.5.10 Cleaning ability - An epoxy coated and a lacquered panel shall be prepared as specified in 4.5.13.1. Observe each panel under an ultraviolet light for flourescence. Replace any panel that shows any sign of fluorescence. Spread 2 drops of oil, conforming to MIL-L-6082, evenly over each panel. Then sift carbon black, conforming to Form I of TT-P-343, through cheesecloth and spread evenly over each panel. Turn the panels over and tap lightly to remove excess carbon. Pour 10 milliliters (ml) of the cleaning compound-distilled water solution specified in 4.5.1 onto each panel and spread evenly over the soiled surface. Place each panel in a horizontal position in a tray. Clean each panel by applying another 10 ml portion of the cleaning compound-distilled water solution and rubbing with absorbent cotton. Flood the panels once again with the cleaning compound-distilled water solution and rinse thoroughly with distilled water. Allow to dry at laboratory room temperature at an angle of 45 degrees from the vertical. Observe the dried panels for the presence of oil as evidenced by irridescence or any increase in the gloss of the panels. Observe the panels under the ultraviolet light for fluorescence resulting from any residual oil on the panels. Repeat this test using the cleaning compound-synthetic hard water solution specified in 4.5.3 with synthetic hard water rinse; and the cleaning compound-synthetic sea water solution specified in 4.5.4 with a synthetic sea water rinse.

4.5.11 Rinsing ability - An epoxy coated and a lacquered panel shall be prepared as specified in 4.5.13.1. Flood each panel with the cleaning compound-distilled water solution specified in 4.5.1 and allow to drain and air dry at an angle of 45 degrees from the vertical. Then place a 250 watt infra-red lamp 1 foot from the panels. After exposing the panels to the heat of the lamp for one hour, rinse the panels thoroughly with distilled water. Observe for any indication of a cleaning compound residue remaining on the panels. Repeat this test using the cleaning compound-synthetic hard water solution specified in 4.5.3 and the cleaning compound-synthetic sea water solution specified in 4.5.4, rinsing respectively with synthetic hard water and synthetic sea water.

4.5.12 Corrosiveness -

4.5.12.1 Test panels - Test panels measuring 0.050 inch by 1 inch by 6 inches shall be prepared in accordance with Table VI.

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TABLE VI

PANEL	SPECIFICATION
Aluminum alloy 2024, polished 1/	QQ-A-250/4
Aluminum alloy, Alclad 2024, polished	QQ-A-250/5
Copper, polished	QQ-C-502
Steel, polished	MIL-S-6721
Magnesium alloy 2/	QQ-M-44
Steel 3/	MIL-S-7952

1/ Panels shall be polished by means of a buffing wheel and buffing compound.

2/ Treated in accordance with Type I of MIL-M-3171.

3/ Cadmium plated in accordance with Type I of QQ-P-416.

4.5.12.2 Procedure - Immerse the six test panels in the cleaning compound-distilled water solution specified in 4.5.1. Do not allow the panels to contact each other. After a one hour immersion, remove the panels and allow to remain flat in a horizontal position for 24 hours. After rinsing thoroughly with distilled water, examine each panel for discoloration, corrosion, chemical attack or any other sign of deterioration.

* 4.5.13 Effect on painted surfaces - An epoxy coated and a lacquered panel shall be prepared as specified in 4.5.13.1. Immerse both panels in the cleaning compound-distilled water solution specified in 4.5.1. After a 45 minute immersion, remove the panels and allow to drain for 5 minutes at an angle of 45 degrees from the vertical. Rinse the panels thoroughly with distilled water and examine the painted surface for softening, blistering, discoloration, cracking or any other sign of deterioration. Repeat this test, immersing a similar set of panels in the cleaning compound-synthetic hard water solution specified in 4.5.3 and in the cleaning compound-synthetic sea water solution specified in 4.5.4.

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4.5.13.1 Painted test panels - Three by six inch panels conforming to QQ-A-250/4, Aluminum Alloy 2024, shall be cleaned by brushing on a 50/50 mixture of cleaning compound, MIL-C-5410 Type II, and Mineral Spirits, TT-T-291 Grade I. Allow the mixture to react with the panels for 15 to 30 minutes and then rinse thoroughly with distilled water. After the panels have dried, apply by brush or dip one coat of chemical film conforming to MIL-C-5541, Class 2. After the chemical film has air dried for 24 hours, spray with one coat of wash primer MIL-C-8514 and air dry 45 minutes to a dry film thickness of 0.0002 to 0.0003 inches. The panels shall then be sprayed with either the epoxy or lacquer system specified in Table VII.

TABLE VII

Coating Epoxy System	Specification	Air Dry Time	Dry Film Thickness (in inches)
Primer	MIL-P-23377	1 hour	0.0005 to 0.0008
White epoxy	MIL-C-22750	1 hour	a thin mist coat
White epoxy	MIL-C-22750	7 hours between coats; final dry of 8 days	0.0014 to 0.0018
Lacquer System			
Primer	MIL-P-7962	45 minutes	0.0003 to 0.0004
Sea blue lacquer	MIL-L-19537	45 minutes between coats; final dry of 24 hrs.	0.0008 to 0.0012

* 4.5.14 Effect on acrylic base plastics - Acrylic plastic conforming to MIL-C-5425 shall be cut into 1 inch by 7 inch by 0.25 inch test specimens. Clean the plastic strips with aliphatic naphtha conforming to TT-N-95 and condition at room temperature for at least 96 hours prior to use. Arrange the test apparatus as shown in figure 1. Set up four plastic strips as Class 1 levers with the fulcrum 2 inches from the clamped end with a load of 5.2 pounds suspended four inches from the fulcrum. This load produces an outer fiber stress of 2000 pounds per square inch at the fulcrum. After 30 minutes and while the strips remain under stress examine for crazing as follows: Place a source of light under the strips and examine the fulcrum area over a 7 power

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magnifier at such an angle that light will be reflected from any cracks or fissures that may be present. Use the light source only during examination to prevent overheating of the plastic strips. Remove any strip that shows any sign of crazing. While the strips are under stress place ten drops of cleaning compound-distilled water solution specified in 4.5.1 at the fulcrum of each of three strips (the fourth is a control). Place a glass slide, 1 by 1.5 inches, over the liquid with its long axis parallel to that of the plastic strip. After the first application of test solution, apply an additional 3 drops of the solution at intervals of 7, 24 and 31 hours to compensate for evaporation losses. Forty-eight hours after the first application of the test solution and while the strips are under stress, examine for crazing as described above. All cracks at the edges of a strip shall be noted as "edge crazing" and disregarded unless they extend across the entire width of the strip. If the control crazes, repeat the test using four different test specimens. Examine the strips also for signs of chemical attack or any other deterioration. Repeat this test using the cleaning compound-synthetic hard water solution specified in 4.5.3 and the cleaning compound-synthetic sea water solution specified in 4.5.4.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging and packing - The packaging and packing of the cleaning compound shall be in accordance with PPP-C-301. Packaging shall be level A or C, packing shall be level A, B or C as specified (see 6.2).

* 5.1.1 Unit container - The cleaning compound shall be packaged in metal cans, pails or drums conforming to PPP-C-301. The type and capacity of the unit container shall be specified by the Procuring Activity (see 6.2).

5.2 Marking - The marking of containers shall be in accordance with MIL-STD-129.

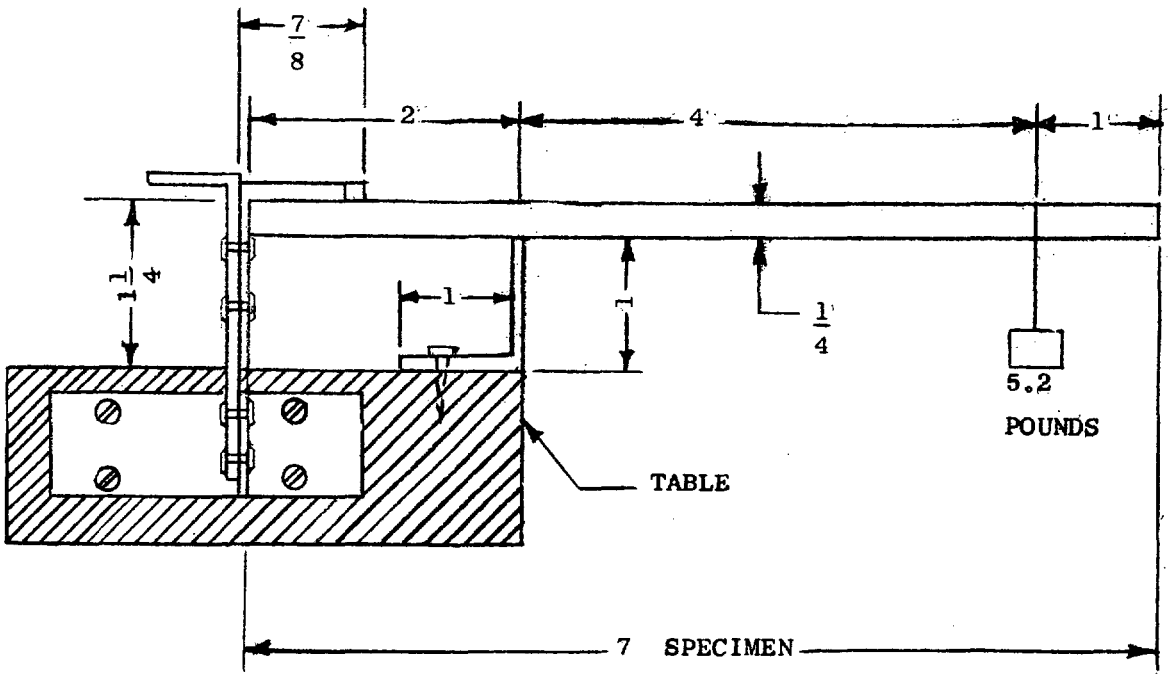
* 5.2.1 Additional marking - Unit containers shall be marked with the following:

(a) Manufacturer's directions for use

(b) Consult NAVWEPS 01-1A-506, aircraft Maintenance Cleaning, for additonal decontaminating procedures.

(c) The lot number and date of manufacture.

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NOTES:

1. DIMENSIONS IN INCHES.
2. 1/8 ANGLE IRON USED AT FULCRUM AND END OF SPECIMEN.

FIGURE 1 - TEST SET-UP FOR CRAZING RESISTANCE OF PLASTICS UNDER STRESS.

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6. NOTES

6.1 Intended use - The cleaning compound, after dissolving in water, hard water or sea water; is intended for use in the decontaminating and cleaning of finished and unfinished aircraft surfaces.

6.2 Ordering data -

(a) Title, number, and date of this specification.

(b) Size and Type of unit container; and quantity (see 5.1.1).

(c) Levels of packaging and packing (see 5.1)

(d) Additional data (see 3.1).

6.3 Marginal notations - The outside margins of this specification have been marked to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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